ETH zürich

Institute for Chemical and Bioengineering

ETH Zurich
Prof. Dr. Gonzalo Guillén-Gosálbez
Vladimir-Prelog-Weg 1, HCI G135
8093 Zurich, Switzerland
Phone: +41 44 633 40 84
gonzalo.guillen.gosalbez@chem.ethz.ch
www.superlab.ethz.ch

Zurich, 17.03.2023

Dr. Michael Rowan
Executive Editor, Royal Society of Chemistry

IT'S A NICE LETTER

Subject: Manuscript submission

Dear Dr. Rowan,

I wouldn't use italics

We are pleased to submit our perspective titled ***"Energy crisis in Europe revamps the economic competitiveness of green chemicals"*** by A. Nabera, I.-R. Istrate, A.J. Martín, J. Pérez-Ramírez and G. Guillén-Gosálbez for consideration in *Green Chemistry*.

The chemical industry is responsible for 10% of global greenhouse gas emissions, despite ongoing efforts to develop low-carbon alternatives. Fossil fuels remain heavily relied upon, particularly in the production of ammonia and methanol, with natural gas dependency accounting for two-thirds of global production. Green hydrogen produced via renewable electricity enable sustainable alternatives, but economic feasibility remains a barrier compared to traditional, fossil resources-based methods.

Despite ongoing economic feasibility challenges, current geopolitical conditions, including surging natural gas and electricity prices, have shifted the cost equation for fossil and green ammonia and methanol production in Europe. Our study found that production costs for fossil ammonia and methanol reached *ca.* 2.4 and 2.2 USD ~~per kilogram~~ kg^{-1}, respectively, during the peak of natural gas spot prices in August 2022, ~~three to four~~ 3-4 times higher than the same period in 2021. Green ammonia and methanol produced through on-shore wind electricity via water electrolysis, however, proved to be economically competitive during the same period. Wind-based ammonia was especially advantageous, with costs ranging from 1.3 to 1.6 USD ~~per kilogram~~ kg^{-1}, outperforming its fossil counterpart. An environmental assessment found that wind-based ammonia not only had a lower carbon footprint but also offered economic competitiveness compared to the fossil fuel route, making it a win-win scenario.

same line

I would use present tense! more actual

makes it lighter

Our study underscores the exceptional potential for Europe to spearhead the shift towards a sustainable chemical industry, considering the proven unpredictability of natural gas supply and price fluctuations. We trust that our study will prompt vigorous discussions within academic, industrial, and policymaking circles to expedite the transition towards ~~more~~ sustainable practices. Please find attached the manuscript, supplementary information and table of content entry.

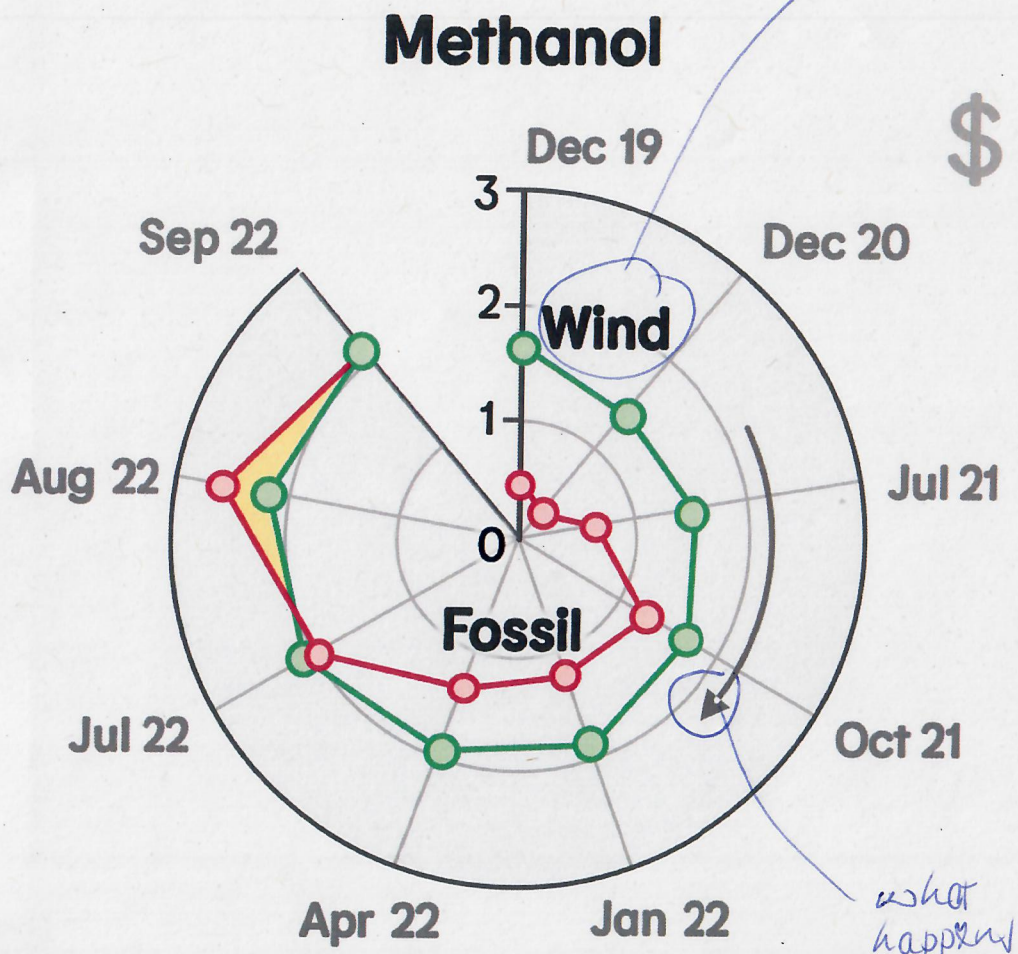
We look forward to hearing from you.

Best wishes,

Prof. Dr. Gonzalo Guillén-Gosálbez on behalf of all the co-authors.

Reviewers!

I would use green instead



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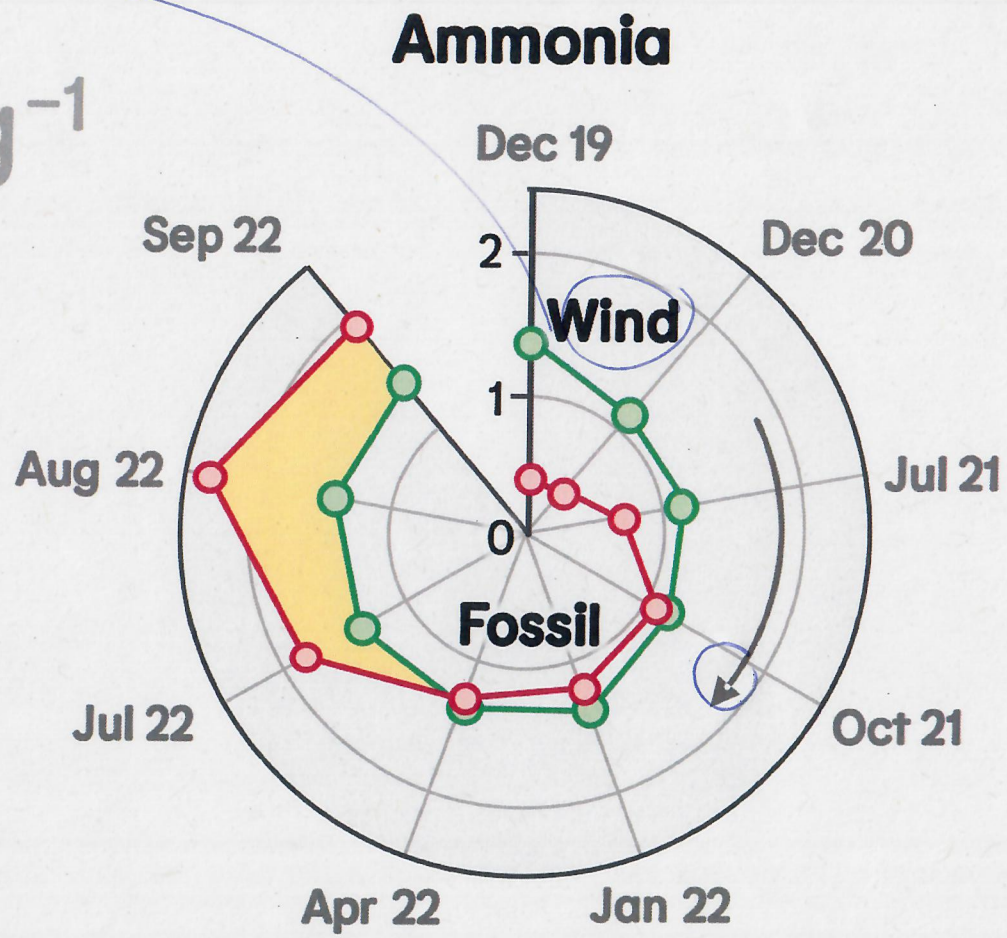

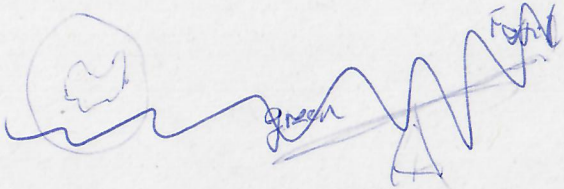

Too large or too? instructions more compact format possible?